ETHAN LARKIN

Software Engineer

Farmersville, TX · larkinethan02@gmail.com · larkin-ethan.github.io

CompTIA Security+ (2025)

ABOUT ME

Associate Software Engineer at L3Harris on the Fusion ELINT team who builds with AI every day: Codex at work, and Claude Code agents at home driven by the specs, guardrails and prompts I write, with tests and CI required on every change. Experienced in C++, Python and Bash, message systems (NATS, protobuf, FlatBuffers), containers and GitLab CI/CD.

EXPERIENCE

L3Harris Greenville, TX

Associate Software Engineer, Fusion ELINT 2025 – Present

- Wrote a C++ daemon that migrated 15 message types from a legacy internal messaging system to NATS and protobuf
- Built GitLab CI/CD pipelines that package the system's ICD (message definitions) as RPMs, replacing a hand-maintained copy in each microservice with a single source installed wherever it's needed
- Expanded test infrastructure with 50+ pytest-bdd scenarios that catch errors before they reach production
- Built 3 web GUIs (HTML, CSS, JavaScript) over existing command-line tools so non-technical users can run them without a terminal, with fewer input errors and no engineer support
- Deploy the subsystem's microservices together with Podman, and built a Kubernetes software set that deploys several analyst tools together
- Designed a standardized FlatBuffers message readable by anyone with the message definition; use SQLite in a self-contained microservice and Bash for subsystem scripting
- Use Codex AI agents to fix bugs, write tests, build tools and review code (est. 40% faster bug fixes); teach teammates

Earlier work (2018 – 2025, through school): QuikTrip, Emerald Cottages, Whataburger

PROJECTS

Drop a Bop — *TypeScript, React, AWS Lambda, DynamoDB, AWS SAM* Fall 2026

- Music-sharing app where friend groups share a song each weekday, rate them, and see results and leaderboards
- Hosted on Cloudflare Pages with a serverless AWS back end (Lambda, API Gateway, DynamoDB, Cognito) defined in AWS SAM; runs at extremely low cost with ~1,000 automated tests
- Built with Claude Code agents from my specs, with tests and CI required; demo at dropabop-demo.pages.dev

Loom — AI resume tailor — *Next.js, TypeScript, Gemini API, Supabase, Stripe* In progress

- Tailors resumes to job postings using only the user's logged history; a system prompt and JSON output per AI call
- Validates AI output in code: each line must cite a real vault entry; personal details stripped before an anonymized AI review

Direction-Finding Drone — *L3Harris-sponsored senior design · C++, CUDA, MATLAB, Jetson Nano* Fall 2024

- Helped design the original FFT processing in MATLAB, then wrote parallel CUDA FFT code in C++ on a Jetson Nano to process SDR-captured signals in near real time
- Led the move to CUDA unified memory to cut compute time, and helped feed a live spectrogram (5 CS + 6 CE students)

EDUCATION

University of Texas at Dallas — B.S. Computer Science *(Richardson, TX)* December 2024

SKILLS

AI	Codex, Claude Code, Gemini API, prompt design, LLM output validation
Languages	C++, C, Python, Bash, CUDA, JavaScript, TypeScript, Java, SQL, MATLAB, HTML/CSS
Messaging & Data	NATS, Protocol Buffers, FlatBuffers, PostgreSQL, SQLite
DevOps & Cloud	GitLab CI/CD, GitHub Actions, Podman, Docker, Kubernetes, RPM, Linux, AWS, Azure
Frameworks & Testing	React, Next.js, Django, pytest-bdd